

After School Elementary Flooding Preparedness Lesson Plans For Germantown, Philadelphia



Haverford and Bryn Mawr College Capstone

Natasha Ring, Jasmin Dluzniewski, and Jo Smith

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Executive Summary

Germantown, Philadelphia is prone to excessive urban flash flooding due to severe storms, which easily overwhelm drainage systems in the city. This capstone project plans four lesson courses for 3-5 graders in the community. Our curriculum focuses on the history of Germantown flooding and environmental racism to shape future disaster resiliency plans surrounding flooding. Utilizing place-based narrative learning and scaffolding techniques to establish an understanding for why Germantown is prone to flooding and what future generations can do to build resiliency in their communities to combat worsening climate change effects.

The paper has been broken up into two main sections, the first which focuses more on the work of designing these lessons and a second which focuses on the actual implementation of these lessons. The first section begins with an introduction to the more general topic of flooding in Germantown and how our group determined that this was the final form that our capstone would take. Reading this section will provide anyone looking to implement these lessons with an understanding of what they can expect from them and how they have been created. We then introduce key teaching theories, such as place-based learning, in the first section of our Literature Review before moving onto a section dedicated to the history of flooding in Germantown. This section is then followed by our Methods & Ethical Framework section, which outlines how decisions were made regarding everything from choosing our topic to the actual design of the lesson plans. Understanding the method behind the madness should enlighten parties interested in implementing these lessons as to how decisions regarding the form of the lesson plans and their development.

The second half of this capstone paper will be more relevant for the actual implementation of these lessons. Here, all four lesson plans and required materials will be included. The first lesson covers the history of flooding in Germantown, followed by a lesson on environmental racism, a third on what to do if it floods, and a final lesson on community care as it relates to flooding. The following two sections contain details targeted to educators from The Water Shed regarding the actual implementation of these lessons, with a final section containing resources and recommendations for implementation. These final two sections are aimed at supporting staff at The Water Shed with how to implement an afterschool lesson plans, and a list of similar programs or local resources who may be beneficial when it comes to ironing out the details of funding and running a program like the one we have designed.

Introduction and Context

For our senior capstone, we have been working with The Water Shed in Germantown, an organization that aims to “empower flood resilience in Germantown through community-driven art, resources, education and science” (“About the Water Shed”). At the beginning of the Semester, all three members of this capstone group selected designing a curriculum for The Water Shed as our final project. Natasha and Jasmin are both Education and Environmental Studies double majors, and selected to be a part of this group due to the overlap between their two majors. Additionally, this gave them an opportunity to consider how concepts learned in Environmental Studies classes could be taught at other levels using the skills gained in Education classes, especially in their placements. Jo chose to be a part of this group due to their belief that

it is important to start environmental education as young as possible, so a curriculum which was open ended as far as the age groups target felt like an interesting project.

We have been designing a series of four lesson plans for 3rd through 5th graders, which we intend to be implemented by The Water Shed. These lessons have been designed for use in an afterschool program made up of 4 sessions lasting roughly an hour each. This after-school program will allow participants to walk away able to answer three main questions in an age-appropriate manner: What is pluvial flooding, what does flooding look like in Germantown, and what infrastructural changes have occurred in Germantown that impact how the area floods?

We settled on this topic and method of instruction after reflecting on our class conversation with Deneene Brockington, The Water Shed's Manager of Programs, early on in the process, where she mentioned that she'd like to see something for elementary school students. It was as a result of these different individuals interests that we decided on creating a series of modular lesson plans for elementary students specifically relating to flooding in Germantown. The specifics of what we are designing and our specific focal questions, however, have been designed by us after a combination of reflection on our conversations with The Water Shed, existing knowledge about teaching students, and research into what a curriculum discussing natural disasters could look like.

Literature Review

Teaching Theories

This section contains concepts that have been identified as key things to consider when writing the lesson plans. We also believe that they are something educators at The Water Shed should be familiar with and keep in mind when adapting these lesson plans for their students.

Scaffolding

Scaffolding can be defined as “an adult controlling those elements of the task that are essentially beyond the learner’s capacity, thus permitting him to concentrate upon and complete only those elements that are within his range of competence” (Puntambekar 2021). This can be done through the use of adapted curricula, technology, physical tools, or some combination of all three. In a classroom, students with 504 plans and Individualized Education Programs (IEPs) may have specific scaffolds outlined; however, educators should be implementing scaffolds into their classrooms for all students regardless of whether or not the accommodations are explicitly outlined (West Virginia Department of Education).

This requires educators to understand each learner as an individual, specifically their strengths and needs as they apply to educational settings, and be able to adjust lessons on the fly to accommodate learners’ needs (Zurek, Torquati & Acar, 2014). Although the lessons included in this capstone are intended to be implemented in an after-school setting rather than a more traditional school setting where teachers have access to these kinds of formalized educational scaffolds, the educators leading these lessons can utilize their experience as educators and knowledge of best practices for teaching students to adjust to the differing needs of each student.

In nature education, scaffolding can be utilized to ensure that activities and information is accessible to all students. These can include low support strategies such as verbal prompts like “let’s count how many storm drains we see” to direct students to a specific feature they are learning about or high support strategies such as identifying each storm drain with a student by having them point at the storm drain at which point the educator names the object (Zurek, Torquati, & Acar 2014). When scaffolds are correctly implemented in classrooms, they give students intellectual independence to explore topics which may have previously been inaccessible to them.

Building intellectual independence through the use of scaffolds is also an example of cultural scaffolding. Cultural scaffolding is a form of Culturally Relevant Pedagogy (CRP), which “includes methods of teaching that connect student experiential knowledge and cultural backgrounds to curriculum as a means of achieving social and educational goals” (Chenowith, 2016). In this capstone project, we will be structuring cultural scaffolding into our lesson plans through explicit connections to Germantown, where the lessons take place, and where students will presumably be familiar with, and through the use of invitations for students to bring in their own experiences to draw connections between the material and their lives outside of a classroom.

Why Afterschool

We chose to write lesson plans for an afterschool program given the wide range of benefits associated with student attendance at well-structured programs. These include academic benefits, such as increased school attendance and long-term

educational success, and life skills such as increased civic-mindedness and a positive work ethic (National Academies of Sciences, Engineering, and Medicine 2024).

We also believed that an afterschool program would be easier for The Water Shed to actually implement, given that adding lessons to a school curriculum could come with additional challenges such as connecting with teachers and logistics surrounding the actual implementation of new curriculum (Nevenglosky 3). This would be especially challenging given that Philadelphia Schools got given a new science curriculum to begin implementing the current 2025-2026 school year, and teachers have expressed frustration surrounding the lack of support in the rollout (Sitrin). Implementing these lessons as an afterschool program at The Water Shed would avoid the added burden of finding local teachers willing and able to implement these lessons in their classrooms.

Additionally, bringing students to a site within their community, such as The Water Shed, reinforces the connection between what they are learning and what they are experiencing in school (“Place-Based Learning and Outdoor Experiences”). This is called place-based learning. Harvard professor Daniel Wilson identifies 4 key ways a place can be utilized in education: learning at a place, where the importance is placed on it being a new place rather than that specific location, learning with a place, where the emphasis is placed on connecting with the people who utilize the space, learning of a place where location is what is being studied, and learning through a place, where systems found at the location is the focus (Nagelhout 2025).

In the case of The Watershed, utilizing place-based learning in their programs will allow them to connect with local community members and serve as a base for

educational programming focused on flood resilience, which they highlight on their website as a priority (“About the Water Shed”) . We aim to use a combination of learning with place and learning of place in our lessons through the focus on connecting with the staff at the Watershed and using the site as a way to learn about pluvial flooding more specifically.

Although the planning required for educators to implement place-based learning, such as building connections with community members and identifying locations, can be high, researchers have found that, when compared to a more traditional lecture-based classroom, there is significantly higher student engagement and understanding of course content (“Place-Based Learning and Outdoor Experiences”). This also helps to support the connections that can be drawn between a student's learning in school and their experiences outside of it, further incentivizing students to continue learning.

Utilizing Narrative-Based Education

Environmental education for children aims to increase environmental awareness through human connection to nature, raising a new generation of environmentalists. Narrative-based education allows children to establish personal connections to their environment and their community, using Germantown Pennsylvania as the community of interest. (Yang 2022), fostering deeper care beyond just remembering environmental facts. This need for environmental conversations will continue to grow as climate change impacts persist worldwide. Students will utilize flooding preparedness as well as develop community care throughout the curriculum. They will learn why Germantown is

prone to flooding, understand flooding risks and develop strategies to prepare the community for flooding events.

Using narrative-based education like histories of flooding in Germantown will be more efficient in teaching young students about flood risks and preparation. Beginning with asking young students what they already know about flooding can open dialogue for narrative-based learning. Yang's study was conducted with students ages 6-8 in 2 elementary schools in China. This age range is slightly younger than our expected after school group consisting of 3-5th graders in America, but this assumes that an older or diverse group of ages may have a more engaging experience compared to the children of the original study. (Yang 2022)

Flooding in Germantown

Our curriculum focuses on the real-life flooding scenarios that occur in Germantown. Beginning with the history of the build structure of Philadelphia, this section explains why Germantown is prone to flooding, what are the common risks associated with pluvial flooding, and how students can become involved in protecting and educating their communities against flooding risks.

History of Flooding

The city of Philadelphia was built on natural wetlands and mudflats, including Eight natural watersheds running through the greater area (Buckner). To keep up with industrial and economic growth, the Pennsylvania Legislature began building the first canal from Philadelphia to Pottsville in 1815. The canal system of the early 1800s allowed for transportation of raw materials (mostly coal and lumber) to be sent further

inland for greater expansion and trade. However the rise of railroad transportation in the 1840s would override canal transportation with speed and less maintenance (Jones). This quick technological turnaround resulted in railroads being built next to canals and slowly phasing out Philadelphia's use of canals until completely abandoning them by the 1970s. Some canals were filled for further industrial development, limiting where stormwater could properly drain. Increased urban expansion meant a loss in the ecological functions of wetlands for the area including carbon storage, wildlife habitats, water filtration, stormwater retention, and of course, flooding protection (Buckner).

Risks

Built over the Wingohocking Creek, Germantown, Philadelphia is particularly prone to inland and urban flash flooding caused by severe and frequent storms and the overwhelmed storm drains of the city. The backup of storm drains paired with the flat surface of Germantown results in a collection of water on the ground's surface called pluvial flooding. This is different from coastal or fluvial flooding caused by storm surge, high tides, and overflowing rivers (Perez). Philadelphia's growing urban landscape full of impermeable surfaces and limited draining infrastructure exacerbates the flooding effects. During storms, paved surfaces and roofs create more runoff that floods sewer system throughout the watershed. This can back up into residential plumbing (Philadelphia Water District). Risks to Germantown residents include flooding of roads, detours, basement flooding, structural damage, and potential water pollution and contamination (Buckner). The cost to repair damages is not an easy task. In February of 2026, the Germantown Masjid Academy was evacuated due to severe flooding

throughout the building. Luckily, students were safe, but were forced to shelter in place in a different location until classrooms were restored. The school reported that they have insurance, but flooding was not covered (Jackson).

Current resources found in the city include the Philadelphia Water District, a government-owned operation as well as The Water Shed, a non-profit in Germantown working with local artist residents to expand knowledge of Germantown flooding (“About the Water Shed”). Supported by the Philadelphia water district, The Water Shed offers resources surrounding disaster planning, education, and flood insurance and protection. Most educational resources are designed for home-owners in the area, with a higher focus on adult education rather than elementary education surrounding flooding. Deneene had mentioned to our class that she was hoping to start flooding education for elementary students at The Water Shed, a third space separate from school and home, so students can learn about Germantown flooding in an open-dialogue low-stakes environment close to home. The Water Shed is a team of scientists, artists and community members, offering a multi-disciplinary atmosphere to cover flooding topics in a variety of ways between narrative-based teaching, place-based learning and hands-on creative activities.

Our proposal suggests The Water Shed as a space for elementary environmental education surrounding flooding directly in the area of interest, Germantown. This will allow young Germantown students to get more involved in their community through an environmental preparedness lens (Yang 2022). Students develop an understanding why Germantown is prone to pluvial flooding and the community risks associated with these events (Yang 2022). Learning how individuals can prepare for future floods will help

build resilience in the Germantown community. Building community resilience around climate change can foster ecological care in communities. The goal of the program is for students to appreciate disaster response techniques to share with others in the Germantown community.

Methods & Ethical Framework

Initial Stages

When starting this project, we knew that despite it being a group project, our capstone needed to be cohesive. This is why our initial steps were communal, and we intentionally refrained from dividing up the work as we developed a stronger plan, choosing instead to continue working on every step as a group. As work progressed and our plan became more fleshed out, we split the work so that we each tackled separate tasks. This allowed us to divide up the work in such a way that we could make more rapid progress, while still utilizing weekly meetings and check-ins via text and comments on our Google Doc to ensure we are on the same page. The frequent check-ins also allowed us to keep things cohesive, something which we identified wanting to ensure was present in early conversations and our team agreement.

Our first step of this assignment was identifying what we wanted the focus of our capstone to be, beyond the basic knowledge that we would be working on a “Germantown Flooding Curriculum”. After reflecting back on our class’s initial conversation with Deneene and the Reverend Chester Williams, we settled on answering two primary questions: What is flooding, and what does flooding look like in Germantown? We settled on these two questions because we wanted to combine a

scientific explanation of what kids are experiencing with their experiences of flooding in Germantown.

This is because there is evidence that children do better in science classes when lessons connect directly to their experiences outside of the school building, and because children aren't too old to start learning flood preparedness (Yang 2022). Additionally, it is not unheard of for students of this age to start learning about other kinds of disasters. Elementary school students throughout the Midwest often participate in tornado drills at school so that they know what to expect if a tornado becomes a danger then, how to recognize the signs of a tornado and what causes a tornado ("Tornado Safety at School - Missouri StormAware"). Hudson, Ohio's Safety Town is another example of programming that prepares students beginning in kindergarten for a wide variety of disasters and dangers that they might encounter in an age-appropriate way. The dangers identified by this program include guns, food allergies, fires, and being a safe pedestrian (Safety Town).

Our next step was to identify what age group we were looking to target. We initially planned on focusing on elementary schoolers in kindergarten through 5th grade, which is what Deneene had mentioned being a good potential age group to focus on given the number of local elementary schools around The Water Shed. The John B. Kelly School, Fidler Academies Plus School, Mastery Charter School at Wister, and the Edward T. Steel School are all located near The Water Shed. Additionally, the Ralph J. Roberts Germantown Boys & Girls Club and the Wissahickon Boys & Girls Club are both located nearby and would be potential places where children would gather.

However, we eventually chose to narrow the scope of grades we were targeting to just 3rd through 5th grade. This decision was made after much discussion, where we realized that having such a wide range of ages could be nice, so that more local students could get involved, we could more effectively structure these lesson plans and accompanying materials if we narrowed our focus. This would also make it easier on whoever from The Water Shed ends up using these materials, given that it would require less adaptation on their end, particularly if they ended up with a very wide range of students in the program.

The next question we needed to tackle was where these lessons would take place and what form they would take. Initially, we had assumed these lessons would happen at The Water Shed as an after-school program (similar to some of the [after school programs](#) available on the City of Philadelphia's website). As a result of the temporary closure of The Water Shed, our group has needed to pivot and change the plan. Now, we plan to provide the lesson plans including many of the necessary materials such as maps and images of flooding to staff at The Water Shed without specifying a location for these plans. This will allow the staff there to make the best decision based on knowledge of their community and existing relationships with families and childcare workers in the area about the best location for the after-school program.

Individual Work

After solidifying our outline for the lesson plans, we finally decided to split up the work. This meant that Jo would be in charge of the Literature Review which is split up into 2 main sections: Teaching Theories and Flooding Information. To find sources, Jo

utilized search terms including, but not limited to, combinations of: Flood, Flooding, Germantown, Philadelphia, Elementary, Disaster, Scaffolding, and Kids on Google Scholar. After identifying potential sources, both scientific studies and news articles, Jo then organized them into the two sections before beginning the Literature Review.

Jasmin took on the responsibility of the lesson plans. Utilizing a template, her primary responsibility of this project is brainstorming and completing (with support as needed) the lesson plans.

Finally, Natasha's primary responsibilities included ensuring that the whole project was cohesive and that any miscellaneous sections including the Methods & Ethical Framework, Recommendations, and the Introduction & Context were completed. Additionally, her role was to write the Teaching Theories section of the Literature Review and assist with the Lesson Plans as needed. All members of the group were responsible for adding to the references section, along with responding to communication about sections (particularly where they were directly connected to others).

All of these steps are being undertaken with the acknowledgement that we are not from the Germantown community. While we can make suggestions based on what we have heard from community members like Deneene and the Reverend, they know their community better than we do. The Water Shed should feel free to adapt our lesson plans to their needs and implement them as they see fit, especially given the limited contact we are able to have with them throughout this process. We worked to express this in our capstone presentation to staff from The Water Shed who were able to attend, and to make it clear that the work done here is what we have envisioned based on our

experiences and research, however we do not know the Germantown community or The Water Shed's capacity in the way that they do. Given this, we expect staff at The Water Shed to adapt lessons as they see fit and in ways that will make the most sense for them to implement.

Lesson Plans

These lesson plans are designed to help guide teaching in a clear, easy and accessible way. Each lesson includes goals, activities and ways to assess student understanding. Lessons usually start with a short introduction of a new topic followed by teaching new ideas, practice activities and a quick review at the end. Materials and suggested activities are included to support different learning needs. These plans are written to be adaptable to fit an after school program with the understanding that each program has different time allowances.

Each lesson plan will be around 1-2 pages and follow the same template. The layout will be simple and organized so teachers can quickly find what they need. In addition to the lesson steps there may be extra materials included such as guiding worksheets, visual aid posters or websites and links to helpful videos when needed. These elements of the lesson plan are meant to support student learning as well as keep students engaged.

LESSON plan

grade: **3-5**

subject: **History of Flooding In GermanTown**

lesson topic: History of Flooding in Germantown

learning objective/s: Explain why flooding happens in cities

- o Flooding happens in cities mainly because rainwater can not soak into the ground the way it does in natural areas. In cities drainage systems can easily get overwhelmed or blocked.

-Identify local waterways by exploring: <https://water.noaa.gov/gauges/phdp1>

-Suggest ways to reduce flooding

materials needed:

- <https://whyy.org/articles/climate-change-flooding-philadelphia/>
- Simple map of Philadelphia
- paper and markers
-

hook/intro:

(5 Minutes) Ask students "what happens to rain when it falls on grass? How about when it falls on the street?"

lesson structure:

ntroduction to key vocab (5-15 minutes) ->

Background: (5-10 minutes)

Long ago Germantown had small streams like the Wingohocking Creek as the city grew people covered the streams to build roads and houses. Water now has less places to go which causes flooding to happen.

Group Activity (15 minutes): In small groups students will be asked to design ways to stop flooding. They can use the papers and markers to draw their ideas.

Prediction activity (15-25 minutes)- Students sit on rug with clipboards and observe different materials, students will predict what will observe water and what will cause run off.

- Materials to predict:
 - o Concrete
 - o Playground surface
 - o Plastic
 - o Soil
 - o Wood
 - o Sponge
 - o Rocks and gravel
- Students can then go outside to test each material with water.
- Ask students to record their results
- Gather students around to discuss observations

Vocab:

Runoff- water that flows over the ground

Drainage- where the water goes and explains the overwhelming of storm drains, the idea that the drains are so full of water there is nowhere for the water to go, causing a "backup" and increases pluvial flooding on the top of earth's surface.

closure: Pluvial flooding- Occurs when extreme rainfall causes overflow within the drainage systems.

One cause of flooding in Germantown ?

LESSON plan

grade: **3-5**

date:

subject: **Environmental Racism**

lesson topic:

- learning objective/s:** Students will be able to define Environmental Racism
- - When Certain groups of people based on their race or income are unfairly affected by environmental problems
 - Explain how flooding affects communities differently
 - Describe flooding in Germantown
 - Identify ways communities can create fair environmental solutions

materials needed:

- Map of Philadelphia
- Images of flooding
- Chart paper or whiteboard
- Markers
- Slideshow with images/videos related to the lesson that I think might be helpful for educators

<https://docs.google.com/presentation/d/1HspfldRPynwl7oPB3MuSK-xz5PM0m68AVc4nPyFPmZE/edit?>

hook/intro:

- Talk for about 10 minutes: "What happens when it rains a lot in your neighborhood?"
 - Think back to the experiment from last session
- Do you think all neighborhoods are affected the same way?

lesson structure:

- Explain:
- Some neighborhoods have more environmental problems than others
 - This can happen because decisions that were made long ago about housing, roads and resources
 - These problems often affect BIPOC communities
- Focus on Germantown flooding:
- Germantown has experienced flooding partly because streams like Wingohocking creeks were covered up and built over
 - Explain why more pavement means less water can soak into the ground
 - Have students research their own neighborhood's environmental challenges

closure:

One example of flooding you have experienced in your neighborhood or a connection to water.

LESSON plan

grade:

date:

subject:

What to do if it floods

lesson topic:

learning objective/s:

- Explain what to do during a flood
- Identify ways to prevent flood damage at home
- Create simple emergency plan

materials needed:

- Flyers from https://www.phila.gov/media/20210128101653/Why_Germantown_Floods_1.22.21.pdf
- Markers and paper
- Water Shed flooding intersections map
- SEPTA app/Google maps
-

hook/intro: What would you do if water started filling your street or home?

- Have students use apple maps to set up alternative routes home
- Mini discussion about what is safe and unsafe during a flood (10-20 minutes)
-

lesson structure:

-Explain and discuss:

- Moving to higher ground immediately
- avoid walking driving through flood water
- Listen to alerts or adults for instruction
- call for help
-

- Preparing your home

- Pick up belongings from the floor
- keep important belongings out of the basemnt
- Know where water might enter your home
- Show the watershed flooding intersections map
- Ask:
- Where are flood-prone areas?
- What places should be avoided during heavy rain?
- Use Google Maps or SEPTA app to:
- Identify safe routes home
- Compare regular vs emergency routes
-

closure:

Students create a simple plan including:

- Safe place to go (home, relative, community center)
- Emergency contact (parent/guardian)
- Two safe routes home
- One thing they would grab quickly
- They can draw or write their plan using markers and paper (10-15 minutes)

LESSON

plan

grade:

date:

subject: Stronger Together: Community Cares During Flooding in Germantown

lesson topic: Community care

learning objective/s:

Students will be able to:

- define community care and explain why it matters during floods
- Describe how flooding impacts neighborhoods like Germantown
- Identify ways people and communities support each other before during and after floods

materials needed:

Poster sized paper
Markers
Paint
Glue

hook/intro:

How can communities work together to stay safe during flooding?

lesson structure:

Warm Up: (10 Minutes):

- Ask students:
 - "If flooding happened in your neighborhood who would you rely on for help?"
 - "How can neighbors help each other?"
 - Write responses on a board
- Introduce community care:
 - Not just individual survival its more about collective responsibility
 - checking on neighbors, sharing food and water , helping clean up after flights
 - Activity:
 - Students will create awariness poster about flooding in their neighborhoods. The students will then do a community walk and hang up the posters in their neighborhood

closure: Why is community care important during disasters?
What is one way you can help your community during a flood?

For Educators From The Water Shed

Deneene, from The Water Shed, expressed a desire to have programming for kids, outside of their past partnership with Mural Arts. This collection of lesson plans have been designed in an attempt to meet that goal, and with the belief that staff at The Water Shed are more than capable of leading this after-school program. These lessons are intended to be an example of one way that they can expand their mission to connect with local school kids through the use of place-based education by doing it in an after-school setting at The Water Shed. Additionally, the after-school setting should allow them to keep it more informal and to run the programming in an environment they are familiar with and with resources they already have.

Staff at The Water Shed who will serve as educators for this program should review the entire document with a focus on the Executive Summary, Literature Review, Lesson Plans, and Recommendations sections. This is especially important if they have not educated children in a formal setting like a sports team or youth group before. Understanding the why behind the different teaching practices used in the lessons will be useful, especially as educators adapt the lessons to best fit the needs of the children participating in the program.

The lesson plans are designed to be implemented in 1-hour long blocks as an after school program, likely offered on the same day every month. This is flexible, and the educators should rely on their professional knowledge of their community members

and what they are comfortable with when scheduling these blocks of time. Required virtual materials like images and slide shows will be provided within the lesson plans, however they may need to be supplemented with materials like markers and paper by educators. For more information about designing and running similar programs, educators should explore the Recommendations section of this capstone or utilize knowledge of similar existing programs.

If staff at The Water Shed are not comfortable leading these lessons, we recommend they reach out to community members who are currently involved in their work for support.

Resources and Recommendations

This section highlights additional resources and recommendations for The Water Shed which may be useful when adapting these lesson plans as needed to best suit their community. It includes a variety of people we recommend they connect with, online resources with examples of similar programs, and concepts which we feel will be useful in how they best shape the program to their needs using their knowledge of the Germantown community.

Expanding the Lesson Plans

After the completion of this Senior Environmental Studies Capstone project, the full document containing all research, lesson plans, and an explanation of the work that has been done over the course of the semester will be shared with The Water Shed. This will allow The Water Shed to understand the thought processes behind the design of the lesson plans, utilization of the limited interactions we have had with staff at The

Water Shed, and research that was done regarding both how the flooding occurs and best practices for educators in the sort of program we have designed. The goal of these lessons was to provide some examples and a basic framework to teach local students about the flooding that impacts them, however, The Water Shed should feel free to modify and expand on these lessons to ensure they are more closely aligned with the needs of specific individuals participating in the programming.

At the beginning of the project, the hope was that The Water Shed staff would be meeting with our group at regular intervals to provide feedback. Deneene Brockington, The Water Shed's Manager of Programs, mentioned that she'd wanted to see The Water Shed do more work with students, but it had never actually come to fruition. Due to a single-vehicle car accident in early March 2026, which caused extensive damage to the building, our communication with staff at The Water Shed was less consistent than initially planned ("About the Water Shed"). This prevented us from working in a more collaborative fashion with Deneene to get more clarity on what she had imagined for The Water Shed's work with students. As a result, we have used knowledge gained from those initial conversations with staff at The Water Shed to guide our thinking, recognizing that this is an imperfect replacement for more sustained collaboration with The Water Shed staff on this project.

We also recommend that The Water Shed Staff reach out to local educators whose students they are planning to target to ensure that it is cohesive with what students may already be learning about flooding and Germantown's history in school. This is also a good chance to connect with community stakeholders since none of us are part of the Germantown Community, something Reverend Chester Williams

emphasized as being a priority for him during our meeting on February 25, 2026. Additionally, these educators will be familiar with strategies that work best for their specific students, something that will help educators at The Water Shed better adapt and scaffold their lessons to students.

Other individuals to whom it may be beneficial for The Water Shed to reach out to ahead of running this afterschool program include parents and guardians of local children. These adults can share feedback about what would work best for their families, including specific procedures for students' arrival and departure from the program and the timing of this program. This is feedback that is specific to the community, and we felt would be important for educators who would eventually implement the program to be aware of.

Additional Resources

If educators are looking to learn more about crafting environmental education lesson plans or how to expand the ones we have developed, we recommend they become familiar with what Meaningful Watershed Educational Experiences (MWEEs) are. [NOAAs Office of Education](#) has many good resources for doing this, but in short a MWEE is “a learner-centered framework that focuses on investigations into local environmental issues and leads informed action” (“What are Meaningful Watershed Educational Experiences (MWEEs)?”). This framework emphasizes the use of experiential education alongside other strategies for learning, such as reading, as a way for learners to interact more directly with their local environment through the use of place-based learning. This helps to improve environmental literacy and connection to

their local environment, which often leads to more sustainable communities (“What are Meaningful Watershed Educational Experiences (MWEEs)?”). Additionally, MWEE learning objectives can, but do not need to, address academic standards in addition to objectives which are applicable outside of a classroom such as civic responsibility and connection to the local environment (“What are Meaningful Watershed Educational Experiences (MWEEs)?”).

There are many good examples of MWEEs covering a wide variety of environments, topics, and target age groups available online. For MWEEs specific to Pennsylvania, we recommend checking out [Sea Grant Pennsylvania](#) whose website contains examples MWEEs done in a variety of environments with different age groups across the state, including students in Philadelphia building model animal habitats. [Pennsylvania Gateway to Green](#) would be another good resource for educators looking to learn more about MWEEs and how they can be implemented, including an Act 45 course for school administrators. If The Water Shed were to be interested in adapting these lessons from an afterschool program, which is what they have initially been designed for, and pivot them towards use in school classroom setting.

[Fairmount Water Works](#), an environmental center located in Philadelphia between Boat House Row and the Art Museum with a focus on educating the public about the importance of water resource protection, would be another great resource to connect with. They have a large number of lesson plans for all grade bands relating to sustainability which have been developed in partnership with local educators over the years available on their page [Understanding the Urban Watershed](#). Additionally, they have been running these school programs for years, sometimes providing transportation

to and from Fairmount Water Works, and their programs are free to all students in the School District of Philadelphia (“School Programs.”). Given that they are associated with the Philadelphia Water Department, like The Water Shed is, they may have useful insight into how to fund these kinds of programs and other additional, relevant resources which are not publicly available.

Educators should also check out our references list since it contains more information about The Water Shed, resources explaining why we have approached this project the way we have, and content that explicitly focuses on topics which we may not have gone into as much depth on. In addition to their in-text citations throughout this capstone, the full citations, often including links, are available within the References section.

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